Decent Coaching Classes

X SB

Mathematics: Part 1 (Ch. 1 - Linear Equations, Ch. 2 - Quadratic Equations) & Part 2 (Ch. 1 - Similarity, Ch. 2 - Pythagoras Theorem)

Time: 1 Hour — Marks: 30 (Answer Key)

Q.1 (A) Choose the correct alternative. (5)

1) b) $ax^2 + bx + c = 0$ $(a \neq 0)$

2) b) no solution

3) a) 2/3

4) a) the sum of squares of the other two sides

5) a) $b^2 - 4ac$

Q.2 (any four) (2 marks each)

1) $x + y = 7$; $x - y = 3$. Adding: $2x = 10$, so $x = 5$. Then $y = 7 - 5 = 2$.

2) $x^2 - 5x + 6 = 0$. Here a=1, b=-5, c=6. $D = 25 - 24 = 1$. $x = (5 \pm 1)/2 \rightarrow x = 3$ or $x = 2$.

3) By BPT: $PX/XQ = PY/YR \rightarrow 2/4 = 3/YR \rightarrow YR = (3 \times 4)/2 = 6$

4) Diagonal of a square = side × $\sqrt{2}$ = $6\sqrt{2}$ cm

5) $2x^2 + 3x + 4 = 0$. $D = 9 - 32 = -23 < 0$. Since $D < 0$, the roots are not real.

Q.3 (any three) (3 marks each)

1) $2x + 3y = 11$ ...(i) $2x - 4y = -24$ ...(ii)

Subtracting (ii) from (i): $7y = 35$, so $y = 5$. Substituting in (i): $2x + 15 = 11$, so $x = -2$.

Check in (ii): $2(-2) - 4(5) = -4 - 20 = -24$ ✓

2) Sum of roots = $4 + (-3) = 1$; Product of roots = $4 \times (-3) = -12$

Required equation: $x^2 - (\text{sum})x + (\text{product}) = 0 \rightarrow x^2 - x - 12 = 0$

3) By Pythagoras theorem: $AC^2 = AB^2 + BC^2 = 5^2 + 12^2 = 25 + 144 = 169$, so AC = 13 cm

Area of $\Delta ABC = 1/2 \times AB \times BC = 1/2 \times 5 \times 12 = 30$ sq. units

4) Statement: If a line parallel to one side of a triangle intersects the other two sides in distinct points, then it divides those sides in the same ratio.

Proof: In ΔABC, seg DE || side BC, D on AB, E on AC. Draw seg DC and seg BE.

$A(\Delta ADE)/A(\Delta DEB) = AD/DB$ ...(equal heights from E on AB)

$A(\Delta ADE)/A(\Delta DEC) = AE/EC$ ...(equal heights from D on AC)

Since ΔDEB and ΔDEC have the same base DE and lie between the same parallel lines DE and BC,

$A(\Delta DEB) = A(\Delta DEC)$. Hence $AD/DB = AE/EC$.

Q.4 (any two) (4 marks each)

1) Let the tens digit be x and the unit digit be y. Number = $10x + y$.

$x + y = 9$ ...(i)

$10x + y + 27 = 10y + x \rightarrow 9x - 9y = -27 \rightarrow x - y = -3$ ...(ii)

Adding (i) and (ii): $2x = 6$, so $x = 3$. Then $y = 9 - 3 = 6$.

The number = $10(3) + 6 = 36$. Check: $36 + 27 = 63$, which is 36 with digits reversed ✓

2) Let Rohan's present age be x years.

$(x - 5)(x + 9) = 15 \rightarrow x^2 + 4x - 45 = 15 \rightarrow x^2 + 4x - 60 = 0$

$(x + 10)(x - 6) = 0 \rightarrow x = -10$ or $x = 6$

Since age cannot be negative, $x = 6$. Rohan's present age is 6 years.

3) By the property of geometric mean (right triangle with altitude to hypotenuse):

$QM^2 = PM \times MR = 9 \times 16 = 144$, so $QM = 12$

$PR = PM + MR = 9 + 16 = 25$

$PQ^2 = PM \times PR = 9 \times 25 = 225$, so $PQ = 15$

$QR^2 = MR \times PR = 16 \times 25 = 400$, so $QR = 20$

(Check: $PQ^2 + QR^2 = 225 + 400 = 625 = 25^2 = PR^2$ ✓)